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The Effects of Online Social Identity Signals on Retailer Demand

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Abstract. Recent shifts in societal discourse have led digital platforms to support equity, inclusivity, and diversity by introducing identity-signaling features, for example, indicators of owner race or gender. In this work, we explore whether, when, and how using those features may impact retailer demand. We tackle this question via a multimethod study. We begin by conducting a controlled experiment on Prolific.co, presenting subjects with actual Google Places business profiles for a set of Black-owned restaurants in Chicago. We randomly vary the presence of Black- and women-owned labels in these profiles and assess subjects' expectations of popularity and quality along various restaurant dimensions. Our results demonstrate that the Black-owned label, in particular, drives significant increases in all outcomes with the effects arising primarily from Black and democratically liberal subjects. Next, we conduct an archival analysis of the effect label adoption has on the physical foot traffic that retailers receive based on SafeGraph's mobile patterns database. Our difference-in-differences estimations yield consistent results; we find evidence of a positive average effect on foot traffic volumes. Further, we show that a rise in visitors from liberal-leaning geographies drives these effects. We discuss implications for digital platform operators and retailers.

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Keywords: social identity • online platforms • marketing • difference-in-differences

1. Introduction

Research in hospitality, economics, marketing, and related disciplines shows that visible signals of proprietor identity, such as shop décor, can influence consumer patronage (Bell et al. 1994, Craig et al. 2008). Numerous digital platforms now enable business owners to convey these signals online, including Google Places, Yelp, TripAdvisor, and Facebook.¹ Here, we investigate the impacts of these digital signals of identity on retailers' demand, focusing on Black and female ownership labels. More formally, we address the following research questions: How does the presence of social identity labels on restaurants' digital storefronts influence consumer decision making? Under what conditions do any such effects manifest and via what mechanisms?

We employ a multimethod analysis. First, we conduct a large, controlled experiment involving 1,200 subjects at Prolific.co, stratified by race and gender. Subjects are shown a set of Google Places business profiles for Chicago restaurants, randomly incorporating Black- and women-owned business labels. We assess subjects' quality perceptions and popularity expectations. We find positive effects for Black-owned business labels,

moderated by subjects' political leanings and demographics. Second, we analyze the adoption of Yelp and Google's Black- and women-owned business labels for a large sample of restaurants in three major U.S. cities. We estimate a series of difference-in-differences (DiD) regressions using SafeGraph's mobile patterns database to quantify label adoption's effect on customer visits. We find positive, statistically significant impacts on visitation volumes, most attributable to consumers from left-leaning geographies.

We establish the robustness of our results in several ways. Notably, we replicate our experimental findings under an economically incentivized design (Heckman et al. 2021) to ensure we elicit truthful responses. Further, we demonstrate the robustness of our archival results to alternative (modern) difference-in-differences estimators (Liu et al. 2023). Broadly, our work contributes to research examining the impacts of online social identity (Taylor et al. 2023). We build on work studying the effect of such signals on off-line activity (Zentner et al. 2013, Gao et al. 2015) and studies on how signals of diversity affect economic outcomes (Mejia and Parker 2021). Our work has practical implications for retailers,

informing them about the potential benefits of adopting these labels, and for digital platform operators as it highlights how labels influence consumer decisions.

2. Related Work

A rich body of literature explores the challenges faced by female and minority business owners. These challenges encompass various aspects, such as resource acquisition (Fairlie and Robb 2007, Fairlie et al. 2022) and consumer biases (Gligor et al. 2021a, b). In the realm of online markets, similar patterns of disadvantage and performance are observed for business proprietors (Ayres et al. 2015, Younkin and Kuppaswamy 2018). Nonetheless, the penalties experienced by minority and female business owners are likely to be context dependent. Under favorable conditions, prominent identification as a female or minority entrepreneur can yield benefits in the marketplace (Greenberg and Mollick 2017, Chan and Wang 2018, Cotropia et al. 2018) for several reasons. For example, in the restaurant industry, factors such as gender, race, and other identities can lead to varying perceptions of quality (Bell et al. 1994, Kovács and Horwitz 2021). Given these tensions, the benefits of social identity labels for retailers remain unclear.

Whereas recent work documents the consequences of online identity information for social interaction (Taylor et al. 2023), relatively little published work examines the impact of identity signals on retailer performance. That said, several recent working papers have emerged on this topic (Kim and Liu 2022, Mitkina et al. 2022, Aneja et al. 2023). Although we are aware of no peer-reviewed, published work that speaks to our research questions, it is nonetheless useful to articulate distinct features of our work relative to these other studies. Our work differs from these works in several notable respects. First, we extend the focus on social identity labels to include gender. Second, our research presents particularly robust, nuanced causal evidence from controlled experiments. Our first experiment captures several different perceptual measures (reflecting subject inferences on different dimensions of quality). Our replication experiment employs an economically incentivized design, which helps ensure that any effects we observe are not driven by demand effects or social desirability biases. This design also allows for an approximation of consumers' willingness to pay in a pseudo-realistic setting.

Our study contributes to the literature more broadly by investigating the effects of social identity labels on consumer decision making and exploring the mechanisms and conditions under which these effects materialize. Our research not only extends the understanding of online signals' influence on off-line outcomes (Zentner et al. 2013, Gao et al. 2015), it also adds to recent work considering how diversity signals may affect economic outcomes (Mejia and Parker 2021). The findings

of our work have practical implications for retailers and digital platform operators, shedding light on the potential benefits of adopting these labels and their influence on consumer choices.

3. Study 1: Controlled Experiment

We begin with an experimental test of the impact of Black- and women-owned labels within Google Places business profiles on consumer willingness to visit the business as well as their expectations of others' interest and their perceptions of the restaurant's quality along various dimensions.

We recruited 1,200 participants from the online research platform [Prolific.co](https://prolific.co) in October and November 2022 over two weeks.² Each subject was paid \$3, translating to an approximate wage of \$12 per hour. To enable exploration of the mechanisms, including racial and gender homophily, we strategically sampled the platform's subject pool to recruit 300 Black and 300 white respondents of each gender. Among respondents, we drop subjects who spend less than 300 seconds responding to the experiment (approximately less than 15 seconds per profile accounting for one minute for pre/post questions) resulting in a final sample of 1,030 subjects; 50.4% of the subjects identified as female, the median age was 35 ($\mu = 37.83$, $SD = 13.72$), and the median reported household income was between \$50,000 and \$75,000. About 72.5% of the respondents identified themselves as politically liberal, whereas the rest as somewhat or largely conservative.

We employed a within-subject, 2×2 factorial design, following the study plan preregistered with the American Economic Association online study registry. In this experiment, each subject was shown 16 randomly selected, hypothetical business profiles (from a pool of 80 different restaurants, bars, and coffee shops based out of Chicago, which were then modified by treatment conditions). For the experiment, we focused on only Google Places profiles and the Black- and woman-owned flags. Each business profile that a subject was shown was randomly assigned to include a woman-owned label (W) (versus not), and each was also independently assigned at random to receive a Black-owned label (B) (versus not). Any given retailer's profile could, thus, receive none, both, or either of the labels in question at random when displayed to a given subject. A particular retailer's profile was shown only once to a given subject, and each subject was shown 16 different retailer profiles. Accordingly, we obtain multiple observations for a given retailer profile under alternative variations of ownership labels. Further, we obtain multiple observations per subject. Given these repeated observations, when performing our analyses, we cluster standard errors by subject and retailer. We also incorporate subject and retailer fixed effects.³

Each business profile that a subject encountered included accurate basic information, including the business name, average review valence, rating volumes, address, hours, contact details, and a graphical depiction of the location on Google Maps.⁴ After viewing each business profile, subjects were asked to evaluate their expected willingness to visit the location (*Self*), their friends' likely willingness to visit (*Others*), and their expectations about the quality of the food (*Food*), service (*Service*), and ambiance/atmosphere (*Atmosphere*), all on a scale from one to five.⁵ The survey also included attention checks toward the end as well as a final request that the subject reports political leaning on a four-point scale from very liberal to very conservative. Scales of this sort are employed in prior studies about political orientation (e.g., Settle et al. 2010). Despite their simplicity, unidimensional, polar scales such as this are shown to be predictive of voting behavior (Jost 2006).

Upon obtaining responses and excluding subjects for inattention, we are left with a sample of 1,030 subjects with a total of 16,480 observations (16 evaluations per subject). To ensure that the resulting sample remains balanced in terms of treatment assignment and other variables of interest, we first assess standardized mean differences in location, age, income, political leaning, gender, and race between all pairs of conditions. All covariates remain balanced; all standardized mean differences less than 0.10, well below the minimum threshold for a "small" effect (Cohen 2013).⁶ That said, to ensure robustness, we also obtain a set of inverse propensity score weights. As can be seen, the reweighted sample exhibits no imbalance. Notably, when we repeat our analyses employing this reweighting scheme, our results (presented next) remain unchanged.

Using this sample (descriptive statistics are in Table C1 in the online appendix), we next estimate a regression of the form expressed in Equation (1). The equation

is indexed by subjects, i , and locations, l . The superscript, ρ , indexes our five outcomes of interest that subjects report. Our equation incorporates a vector of location and subject fixed effects, accounting for subject-specific variation in responses, given our within-subject design. Finally, we also control for the order in which a location l is presented to subject i , $QuestionSeq_{i,l}$:

$$Outcome_{i,l}^{\rho} = \beta \cdot BlackOwned_{i,l} + \eta \cdot WomenOwned_{i,l} + \nu \cdot BlackOwned_{i,l} \cdot WomenOwned_{i,l} + \delta \cdot QuestionSeq_{i,l} + \alpha_i + \lambda_l + \mu_{i,l}. \quad (1)$$

The results of this estimation are presented in Table 1. On average, we observe no significant effects of the woman-owned business label with the exception of a marginally significant, positive effect on expectations of service quality ($p < 0.10$). In contrast, we find that the Black-owned business label drives positive effects on nearly every outcome. In particular, we observe statistically significant, positive effects on subjects' average willingness to visit the location ($p < 0.05$) as well as their expectation that they would enjoy the atmosphere ($p < 0.01$). Further, we observe positive, marginally significant effects on subjects' expectations of food and service quality ($p < 0.10$). Finally, it is worth noting that we observe no evidence of an interaction between the Black- and woman-owned business labels as all coefficients are statistically insignificant.

We next consider heterogeneity in the effect of the two labels by subject's political leaning and demographics. Recent research observes that marketers' and businesses' expression of support for sociopolitical movements has consequences for consumer and employee engagement (Burbano 2021, Wang et al. 2022).⁷

For the sake of interpretation, we conduct these estimations separately for each label. We begin with the Black-owned label, estimating an equation of the form expressed in Equation (2). We consider three moderators,

Table 1. Impact of Google's Black and Women-Owned Labels on Subjects' Rating of Outcomes (Ordinary Least Squares)

DV = 1–5 rating	Self	Others	Food	Service	Atmosphere
Black-owned	0.049* (0.024)	0.033+ (0.020)	0.038+ (0.020)	0.031+ (0.017)	0.051** (0.019)
Women-owned	0.016 (0.024)	0.012 (0.017)	0.002 (0.020)	0.034+ (0.018)	0.015 (0.022)
Black- and women-owned	0.015 (0.033)	0.017 (0.026)	0.009 (0.031)	0.010 (0.026)	0.020 (0.031)
Question sequence	−0.004 (0.007)	−0.005 (0.007)	0.005 (0.006)	0.011+ (0.006)	0.001 (0.008)
Number of observations	16,480	16,480	16,480	16,480	16,480
Adjusted R^2 (full model)	0.333	0.354	0.326	0.376	0.332
Adjusted R^2 (proj model)	0.001	0.000	0.000	0.001	0.001
Location fixed effects	Yes	Yes	Yes	Yes	Yes
Subject fixed effects	Yes	Yes	Yes	Yes	Yes

Notes. Standard errors are clustered by subject and retailer. Expected willingness to visit the location (*Self*), their friends' likely willingness to visit (*Others*), and their expectations about the quality of the food (*Food*), service (*Service*), and ambiance/atmosphere (*Atmosphere*), all on a scale from one to five.

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$; + $p < 0.1$.

namely, subject race, gender, and political leaning. Subsequently, we perform the same estimation replacing the Black-owned label dummy with a woman-owned label dummy:⁸

$$\begin{aligned} Outcome_{i,l}^p = & \beta \cdot BlackOwned_{i,l} + \eta \cdot BlackOwned_{i,l} \\ & \cdot ConservativeSubject_{i,l} + \nu \cdot BlackOwned_{i,l} \\ & \cdot BlackSubject_{i,l} + \gamma \cdot BlackOwned_{i,l} \\ & \cdot FemaleSubject_{i,l} + \delta \cdot QuestionSeq_{i,l} \\ & + \alpha_i + \lambda_l + \mu_{i,l}. \end{aligned} \quad (2)$$

The results of these estimations are presented in Tables 2 and 3. Broadly, we observe no statistically significant sources of moderation around the woman-owned label. However, in Table 2, we observe a great deal of significant moderation around the Black-owned label. First, we see that the effect of the Black-owned label is negatively moderated by conservatism.⁹ Further, the effect of the label is significantly more positive among Black and female subjects. These results reflect the importance of political preferences and implicit support for the underlying social movements that led to the introduction of these labels. Further, these results reflect the importance of both homophily and, potentially, solidarity among marginalized groups as drivers of positive response.

To establish the robustness of these results and ensure they are not driven by demand effects or social desirability biases, we conduct a replication (reported in Online Appendix D), wherein we adopt an economically incentivized design and revisit the effects of the Black-owned business label. We recruit 600 subjects, allocate each subject \$30, and ask that they allocate the funds between a random draw of three restaurants,

which are, in turn, randomly allocated the Black-owned business label. We observe there that the Black-owned business label results in an estimated 6.5% increase in funds that consumers allocate toward a restaurant.

The results of these experiments provide an initial indication of whether and which consumers are likely to respond to these labels. The experiment demonstrates that the labels induce a desire to visit the locations among some consumer groups. Further, the results demonstrate that the labels induce positive quality inferences.

4. Study 2: Archival Study

Whereas our experimental analyses provide causal evidence of race and gender labels' influence on consumer perceptions, generalizability may remain a question. Accordingly, we next address these labels' impact on real consumer behavior by conducting a second study, using archival data from Yelp and Google, capturing actual businesses' adoption of labels, in tandem with third-party data from SafeGraph, capturing consumer foot traffic to specific retailers (described subsequently). We analyze the impacts on demand that retailers experienced following label adoption, considering businesses located in three major U.S. cities.

4.1. Data and Context

The various identity labels were introduced by each platform at different points in time. However, the timing of these introductions is likely to have been driven in part by shifts in societal discourse, triggered by notable events (e.g., the death of George Floyd and the Black Lives Matter (BLM) protests that followed). This raises a concern for causal inference because the effects of adopting a social identity label may be confounded by parallel changes in demand arising from shifts in the attention

Table 2. Heterogeneity in Impact of Black-Owned Label

DV = 1–5 rating	Self	Others	Food	Service	Atmosphere
Black-owned	0.022 (0.037)	0.016 (0.036)	–0.005 (0.034)	0.023 (0.026)	0.075 ⁺ (0.039)
Black-owned · conservative subject	–0.045* (0.018)	–0.037* (0.015)	–0.035* (0.015)	–0.050*** (0.014)	–0.064** (0.022)
Black-owned · Black subject	0.085* (0.033)	0.061 ⁺ (0.031)	0.097** (0.029)	0.067** (0.024)	0.049 (0.031)
Black-owned · female subject	0.072* (0.034)	0.063 ⁺ (0.031)	0.064* (0.031)	0.058* (0.025)	0.052 ⁺ (0.030)
Question sequence	–0.004 (0.007)	–0.005 (0.007)	0.005 (0.006)	0.011 ⁺ (0.006)	0.001 (0.008)
Number of observations	16,480	16,480	16,480	16,480	16,480
Adjusted R ² (full model)	0.334	0.355	0.327	0.376	0.333
Adjusted R ² (proj model)	0.002	0.001	0.002	0.002	0.002
Location fixed effects	Yes	Yes	Yes	Yes	Yes
Subject fixed effects	Yes	Yes	Yes	Yes	Yes

Notes. Standard errors are clustered by subject and retailer. Expected willingness to visit the location (*Self*), their friends' likely willingness to visit (*Others*), and their expectations about the quality of the food (*Food*), service (*Service*), and ambiance/atmosphere (*Atmosphere*), all on a scale from one to five.

⁺*p* < 0.1; **p* < 0.05; ***p* < 0.01; ****p* < 0.001.

Table 3. Heterogeneity in Impact of Woman-Owned Label

DV = 1–5 rating	Self	Others	Food	Service	Atmosphere
Women-owned	0.025 (0.042)	0.018 (0.028)	−0.025 (0.039)	0.049* (0.023)	0.030 (0.030)
Women-owned · conservative subject	−0.012 (0.021)	−0.014 (0.014)	−0.001 (0.018)	−0.026+ (0.014)	−0.033+ (0.018)
Women-owned · Black subject	0.016 (0.034)	0.020 (0.027)	0.052 (0.033)	0.038 (0.028)	0.046 (0.030)
Women-owned · female subject	0.004 (0.036)	0.015 (0.030)	0.012 (0.031)	−0.006 (0.025)	0.010 (0.030)
Question sequence	−0.004 (0.007)	−0.005 (0.007)	0.005 (0.006)	0.011+ (0.006)	0.001 (0.008)
Number of observations	16,480	16,480	16,480	16,480	16,480
Adjusted R^2 (full model)	0.333	0.354	0.326	0.375	0.331
Adjusted R^2 (proj model)	−0.000	−0.000	−0.000	0.001	0.000
Location fixed effects	Yes	Yes	Yes	Yes	Yes
Subject fixed effects	Yes	Yes	Yes	Yes	Yes

Notes. Standard errors are clustered by subject and retailer. Expected willingness to visit the location (*Self*), their friends' likely willingness to visit (*Others*), and their expectations about the quality of the food (*Food*), service (*Service*), and ambiance/atmosphere (*Atmosphere*), all on a scale from one to five.

+ $p < 0.1$; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

consumers were paying to business operators' social identities and popular support for minority businesses in particular. To address this concern, our focal estimations employ a constrained sample of businesses that were owned and operated by individuals who had the ability to adopt a particular label, that is, a set of verified Black-owned businesses and a set of verified female-owned businesses. These narrower samples provide an additional benefit; they ensure that our estimated causal effects are based upon a valid counterfactual for label adoption. Were we to contrast label adoption with nonadoption in the broader restaurant population, our estimates would be comparing adopting Black-owned businesses (for example) with nonadopting Black-owned businesses and with businesses owned by other demographics (e.g., white-owned businesses).

Our indicators of restaurants' label adoptions are obtained from Google Places and Yelp. Our measures of consumer demand are obtained from SafeGraph. SafeGraph provides geospatial data derived from data shared by a large set of mobile app partners, which record users' location information. Using geofencing, location data are aggregated by SafeGraph to create monthly/weekly estimates of unique visitors, visits, and median dwell times at myriad physical locations, including retail spaces and public buildings. Brelsford et al. (2022) provide more details about SafeGraph data and report on empirical validation of SafeGraph measures.¹⁰ Several recent studies notably use SafeGraph data to understand policy effects on individual mobility patterns (e.g., Carranza et al. 2022, Shin et al. 2022).

We obtained label adoptions for a large sample of coffee shops, restaurants, and bars located in Boston, Chicago, and Memphis.¹¹ In each city, businesses were identified on Google Maps via searches employing the

terms "restaurant," "coffeeshop," and "bar." After identifying the presence (or absence) of various social identity labels, we identified the corresponding listings for each establishment on Yelp and, again, recorded whether those profiles bore any social identity labels. Next, we matched the sample of businesses to SafeGraph place keys based on location name and address. This matching enabled us to construct panels of monthly foot traffic for each business from January 2018 through April 2022. Finally, we exclude from this panel any businesses for which any monthly values were missing or that closed prior to the end of the observation window. We are left with 2,965 (2,410 independent) businesses in Chicago, 1,746 in Boston (1,277 independent), and 1,564 in Memphis (934 independent). Our final sample notably omits chain locations as we anticipate that a strong brand is likely to overpower or muddle the consequences of proprietor social identity.¹² The descriptive statistics for visits and adoption of each label are provided in Table 4.

4.2. Coding of Verified Owner Demographics

Although we begin with an analysis of Black- and woman-owned labels' effects on the broader sample of retailers in our data set, our focal estimations are ultimately based on a restricted sample of verified Black- and female-owned businesses. By focusing on these subsamples of coder-verified owner demographics, we can establish a meaningful counterfactual, and we simultaneously rule out the potential that the estimated effects are confounded by other systematic differences in business performance across owner demographics. For example, because Black-owned business labels were introduced shortly after the death of George Floyd and the rise of the BLM movement, were we to ignore owner race in the control group, the effect of the Black-owned

Table 4. Descriptive Statistics for the DiD Indicators and Primary Dependent Variable in the Monthly Visit Panel ($N = 219,200$)

Variable	Date	Mean	Standard deviation	Minimum	Maximum	Percentage of total places that adopt	Percentage of verified ^a identity places that adopt
Google Black-owned	2020-07-31	0.0199	0.14	0.00	1.00	4.8	50.9
Google woman-owned	2018-04-07	0.231	0.421	0.00	1.00	24.6	50.7
Google LGBTQ friendly	2018-06-28	0.205	0.403	0.00	1.00	22.5	—
Google veteran-led	2018-08-27	0.018	0.133	0.00	1.00	2.2	—
Yelp Black-owned	2020-06-18	0.00748	0.0861	0.00	1.00	1.7	18
Yelp woman-owned	2020-02-27	0.0532	0.224	0.00	1.00	10.3	13.8
Yelp Asian-owned	2021-04-06	0.00778	0.0879	0.00	1.00	3.2	10.9
Yelp LatinX-owned	2020-09-15	0.016	0.126	0.00	1.00	4.2	21.8
Visits	—	210.26	1,224.25	1.00	274,483	—	—

Note. Dates indicate when labels became searchable as a result of Google/Yelp policy change.

^aVerified identity for the Black-owned labels implies across the three cities and for the rest of the labels only for Chicago.

label adoption may be confounded by broad increases in demand for Black-owned businesses generally, not simply for those that adopted the label. The same concern might arise for the woman-owned label in relation to the #MeToo movement and other social movements related to the other social identity labels; for example, the Asian-owned label on Yelp was introduced following the “stop AAPA hate” movement. Our demographic-specific subsample estimations specifically address this concern by conditioning on a particular owner demographic and contrasting label adopters against proprietors from the same group who did not adopt the relevant label.

Our verified sample of Black-owned establishments spans all three cities and comprises 438 businesses, identified based on newspaper listings, listicles, blogs, and other online resources. To identify our sample of verified female-owned businesses, we found this more difficult to discern. Accordingly, we employed three undergraduate students to assist us in manually verifying owner demographic information for several hundred retailers in our Chicago sample.¹³ To ensure the coders properly understood the instructions, they were first instructed to gather the relevant information for a pilot set of 25 restaurants and to judge the gender and racial demographics of associated owners. Upon demonstrating 90%+ agreement on the various pieces of information collected, the coders were then asked to collect data on an additional 250 restaurants each. A list of 764 independent establishments was ultimately coded in the Chicago area, for which owner demographic information was readily apparent. Note that these establishments have verified owner demographics, but only a fraction of them adopted Black- or woman-owned business labels on Google or Yelp; 63% of Black-verified establishments adopt either the Google or Yelp Black-owned label, and 53% of female-verified establishments adopt either the Google or Yelp female-owned label.

4.3. Econometric Specification

Our key dependent variable in these analyses is the monthly count of unique visits (i.e., foot traffic) to a particular establishment as recorded by SafeGraph. We begin our analyses by considering the effects of the Black- and woman-owned business labels in our entire sample, spanning the three cities of interest. Our empirical specification is expressed by Equation (3), where i indexes individual places of business (restaurants, coffee shops, and bars) and t indexes time (i.e., unique year-months).

$$Visits_{i,t} = \Sigma_{\rho} Label_{i,t}^{\rho} + \alpha_i + City_i \cdot \tau_t + \epsilon_{i,t}. \quad (3)$$

Here, $\Sigma_{\rho} Label_{i,t}^{\rho}$ reflects a set of binary treatment indicators capturing the adoption of various social identity labels. These treatments are set to one if restaurant i eventually adopted label ρ by the end of our panel and if the observation in question took place at a point in time, t , following the date of label ρ 's introduction by its host platform. We construct individual labels in this manner for the entire set of $j \in [GoogleBlackOwnd, GoogleWomenLed, GoogleVeteranOwnd, GoogleLGBTQFrndly, YelpBlackOwnd, YelpWomenOwnd, YelpAsianOwnd, YelpLatinXOwnd]$, defining them independently, depending on whether a business ultimately adopted each and depending on the timing of the particular label's initial introduction onto its respective platform, that is, Google or Yelp. Each label indicator is, thus, equivalent to the interaction between an indicator of treatment and an indicator of the postperiod as in a standard DiD specification.

It is worth noting that the measurement error inherent in our definition of the treatment is strictly one-sided in that our treatment indicator is coded as one from the outset of the feature introduction despite that, in many cases, the retailer might have adopted the label only after a delay. Fortunately, recent theoretical work notes that this sort of measurement error in a treatment definition under a DiD framework results only in attenuation bias; that is, treatment effect estimates are biased

toward zero (Lewbel 2007, Denteh and Kédagni 2022). Further, other work observes that a large majority of Yelp’s Black-owned label adopters did so within one month of the label’s introduction (Aneja et al. 2023). Accordingly, our estimates of treatment effects are likely to be understated and, thus, conservative.

We incorporate restaurant fixed effects, α , to control for any systematic cross-sectional differences between restaurants. Further, we include city-year-month (time) fixed effects, $City \cdot \tau$, to account for market-specific (city-specific) shocks to consumer demand experienced by all establishments in a city.¹⁴ Finally, ϵ is an idiosyncratic error term. The descriptive statistics for the binary DiD indicators are shown in Table 4.

4.4. Results

In Table 5, column (1), we begin by reporting the results of our estimation based on the complete sample of all retailers across the three cities.

This result indicates the effect of label adoption on retailer foot traffic relative to all nonadopting retailers. It should be noted here, however, that among retailers who did not adopt are shops operated by individuals who, in theory, are incapable of adopting; for example, a white-owned restaurant would not be able to adopt a Black-owned business label. Although some media reports point to retailers caught adopting the Black-owned label disingenuously,¹⁵ by and large, the labels are likely adopted only on a legitimate basis.

Considering the results in column (1), we see no evidence that the Black- or woman-owned business labels significantly influence retailer demand though we do see an indication that the Yelp Asian- ($\beta = 27.01$; $p < 0.05$) and Yelp LatinX-owned ($\beta = 34.04$; $p < 0.01$) labels may drive positive response. Turning attention to our focal estimations, we begin with the verified sample of Black-owned businesses for which results are reported in column (2). We see here very clear, positive, significant effects of Black-owned label adoption on both Yelp and Google Places.

Considering column (6), which repeats our estimation on the sample of verified Chicago-area female-owned businesses, once again, we see a positive, statistically significant effect of the Yelp woman-owned label. We do not, however, observe any significant effect from Google’s woman-owned business label (note that there are very few pretreatment observations for this label). To gain a sense of treatment effect dynamics, we also consider estimated effects over a shorter window around the treatment event, limiting our samples to six months before and after. The associated results are shown in columns (3) and (7) for the Black- and female-verified samples, respectively.

We also consider possible systematic differences between Black- and female-owned businesses that did versus did not adopt their respective labels in terms of various observable characteristics. We calculate propensity scores based on a first stage logistic regression, and we use them, in turn, to adjust for imbalance, implementing

Table 5. Impact of Social Identity Labels on Monthly Visits to Independent Restaurants

	Full	Black-verified				Female-verified			
		Unadjusted		Adjusted		Unadjusted		Adjusted	
	Full (1)	Full (2)	Six months (3)	Full (4)	Six months (5)	Full (6)	Six months (7)	Full (8)	Six months (9)
Yelp Black-owned	23.97 (17.05)	50.03** (17.03)	15.81* (7.24)	53.76** (20.00)	13.63+ (8.22)	46.59 (37.56)	26.18 (19.59)	63.90 (43.24)	37.92+ (20.01)
Google Black-owned	1.34 (21.50)	45.20*** (13.12)	21.14*** (6.31)	14.54 (19.06)	13.00* (6.11)	−35.72 (45.38)	−16.67 (33.66)	−14.54 (43.09)	−17.65 (39.99)
Yelp women-owned	14.34 (15.60)	−24.04 (22.53)	−0.507 (16.54)	−20.34 (19.75)	0.13 (14.75)	44.61* (21.21)	59.89*** (16.30)	34.55 (22.88)	60.54*** (16.81)
Google women-owned	−14.61 (13.67)	−16.41+ (8.54)		−10.67 (9.31)		−16.93 (12.16)		−9.80 (14.39)	
Yelp LatinX-owned	34.04** (11.32)	−23.73 (36.02)	−21.66 (25.25)	−30.13 (38.22)	−25.72 (28.36)	47.81* (21.83)		63.33*** (16.49)	
Yelp Asian-owned	27.01* (13.50)					−7.57 (18.13)		0.04 (18.75)	
Google veteran-led	11.84 (16.96)	25.84+ (13.62)		30.56* (14.75)		−18.58 (45.61)		−8.32 (41.48)	
Google LGBTQ+ friendly	−24.23 (22.82)	16.97 (10.57)		16.47 (13.26)		−7.45 (17.00)		−10.92 (15.39)	
Number of observations	219,200	20,964	5,256	20,964	5,256	11,700	2,925	11,700	2,925
Adjusted R^2 (full model)	0.402	0.623	0.785	0.482	0.779	0.620	0.694	0.644	0.694
Adjusted R^2 (proj model)	0.000	0.011	0.006	0.003	0.003	0.009	0.009	0.008	0.007
Fixed effects	Place and city-by-month-year (month-year in the case of female-verified as sample is Chicago only)								

Notes. Standard errors clustered by retailer. The adjusted samples consist of the particular identity-verified samples each propensity weighted by the propensity of adopting either of Yelp/Google’s labels corresponding to the relevant identity.

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$; + $p < 0.1$.

inverse propensity for treatment weighting. Figure E1 in Online Appendix E presents a graphic depiction of mean differences between adopters and nonadopters before and after adjustment in terms of various restaurant and location characteristics. Repeating all of our earlier estimations using the adjusted samples, we obtain the results reported in columns (4) and (5) and (8) and (9) in Table 5. All results remain consistent.¹⁶

4.4.1. Event Study Estimation. Next, we implement a dynamic variant of our main regression specifications, otherwise known as an event study estimation. This estimation is a standard approach to evaluating the assumption of parallel trends and examining the post-treatment dynamics of any effects. In the case of the Black-owned label, we leverage our entire sample of identified Black-owned businesses across all three cities. In the case of the women-owned label, we leverage our Chicago-based sample of manually verified female-owned businesses. We separately consider the adoption of the Black-owned labels in the Black-owned restaurant sample and the adoption of the women-owned labels in the female-owned restaurant sample.

In the case of the Black-owned business labels introduced by Google and Yelp, it is important to note that they were introduced in rapid succession. As noted earlier, Google introduced its Black-owned business label in June 2020, and Yelp introduced its label in July 2020. Given the close proximity of each treatment event, we consider these two labels jointly, defining treatment for the purposes of the event study estimation as a restaurant's ultimate adoption of either the Google or Yelp Black-owned business label. Note, however, that we also report event study estimations that consider each label's effects separately in Online Appendix E, Figure E2. In the case of the women-owned business label, we focus our attention on the Yelp label because Google's label was introduced very early on, in March of 2018, such that we have just three observations prior to treatment onset. Again, however, we report the estimations associated with the Google women-owned label in Online Appendix E, Figure E2.

Our estimation approach is similar to that described by Donnelly et al. (2021). For the Black-owned restaurant sample, we estimate a regression of the form expressed by Equation (4), and for the female-owned sample, we estimate a regression of the form expressed by Equation (5):

$$\text{Visits}_{i,t} = \tau_t + \tau_t \cdot \text{BlackOwnedLabel}_i^{\text{Yelp|Google}} + \sum \text{OtherLabels}_{i,t} + \alpha_i + \epsilon_{i,t}, \quad (4)$$

$$\text{Visits}_{i,t} = \tau_t + \tau_t \cdot \text{WomenOwnedLabel}_i^{\text{Yelp}} + \sum \text{OtherLabels}_{i,t} + \alpha_i + \epsilon_{i,t}. \quad (5)$$

In Equation (4), we define our treatment as those restaurants that eventually adopted either the Google or Yelp

Black-owned business label, $\text{BlackOwnedLabel}_i^{\text{Yelp|Google}}$. We interact this treatment indicator with a vector of year-month dummies, τ_t , omitting May 2020, the month just prior to Yelp's (slightly earlier) label introduction. Similarly, in Equation (5), we define our treatment group as those restaurants that eventually adopted Yelp's women-owned business label, $\text{WomenOwnedLabel}_i^{\text{Yelp}}$. Again, we interact the treatment indicator with a vector of year-month dummies, τ_t , omitting February 2020, the month just prior to Yelp's label introduction.

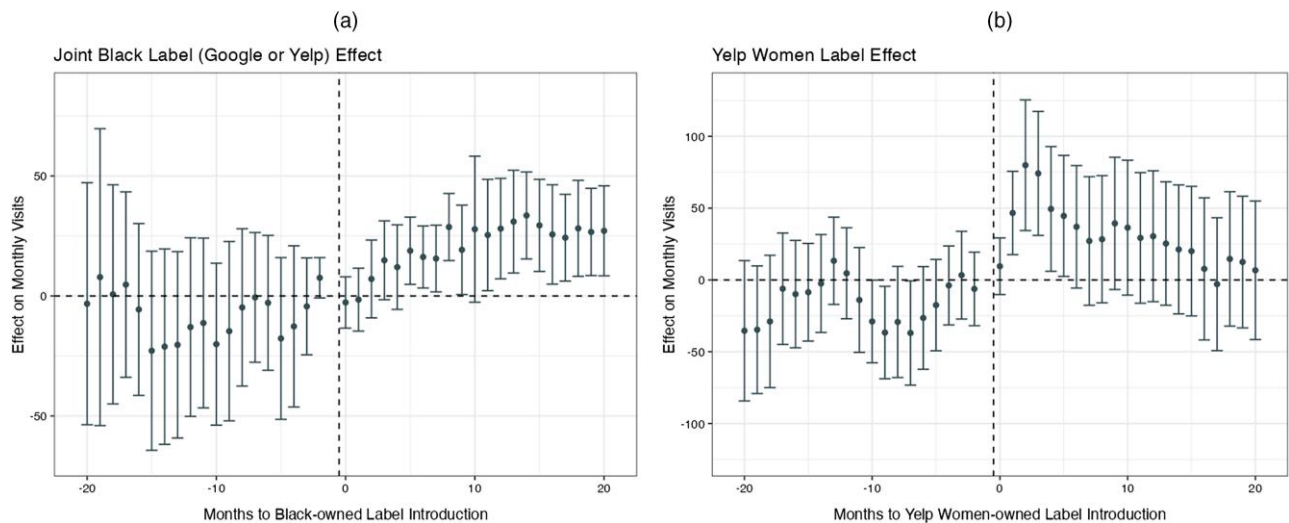
In each regression, we include fixed effects for restaurants, and we control for time-varying binary indicators reflecting each restaurant's adoption of any other social identity labels we have recorded in our data as noted in Table 4. These indicators are set to one beginning in the month of each label's introduction for those restaurants that ultimately adopted them. For example, when we estimate Equation (5), in addition to our estimation of the interaction between year-month dummies and the adoption indicator for Yelp's woman-owned label, we also control for indicators of restaurants' adoption of the Google women-owned label, the Google Black-owned label, the Yelp Black-owned label, the Google veteran-led label, the Google LGBTQ+ friendly label, the Yelp Asian-owned label, and the Yelp LatinX-owned label. Finally, in each regression, we cluster standard errors by restaurants.

For the sake of brevity, we present the estimation results graphically. Figure 1(a) and (b), presents the estimates associated with each specification. In each case, we observe no evidence of significant differences in the trend of visit volumes between treated and untreated restaurants in the lead-up to either label introduction. At the same time, we observe that adopting restaurants exhibit statistically significant increases in their visit volumes after label introduction and adoption. These results indicate no evidence that the parallel trends assumption has been violated, and they further suggest that our estimates are causal.

We also estimate the Liu et al. (2023) counterfactual estimators (shown in Online Appendix E.1), which use factor-augmented models and matrix completion (Athey et al. 2021) to impute counterfactual outcomes for treated panels and thereby estimate an average treatment effect on the treated. Our results are largely consistent, and the placebo equivalence tests indicate no significant pretreatment trends.

4.4.2. Heterogeneity in Visitor Characteristics. Finally, we examine heterogeneity in the demographics and political leaning of visitors arriving at a retailer following label adoption, similar to the analyses we conducted in our experiment. In contrast to the experiment setting, we do not observe visitor demographics directly; SafeGraph provides information on the number of visitors arriving at a particular retailer in a given month who

Figure 1. Event Study Plots for Dynamic Difference-in-Differences



Notes. (a) Any Black-owned (Google or Yelp) label. May 2020 is omitted as reference. (b) Yelp women-owned label. February 2020 is omitted as reference.

maintain their residence in each census block group (CBG). Using CBG demographic estimates from the 2020 American Community Survey,¹⁷ we are, thus, able to determine whether visitors reside in a majority Black or white neighborhood. Similarly, leveraging CBG-level voting estimates for elections conducted in 2021 (Voting and Election Science Team 2022), calculated based on the method of Amos et al. (2017), we can discern whether visitors reside in majority Democrat or Republican neighborhoods. We construct monthly measures, per place, of visitors arriving from Black versus white majority neighborhoods and from Democrat versus Republican majority neighborhoods. Recognizing that the approach we take to aggregating visitors by home-CBG characteristics requires some arbitrary choices, we consider three alternative measures. First, we consider the estimated count of visitors per demographic by multiplying the visits from the CBG by the demographic percentage. Next, we consider the count of visitors arriving from CBGs that exhibit a simple majority of Black/white/Democrat/Republican residents. Third, we consider the count of visitors arriving from CBGs that exhibit a supermajority of residents that belong to the relevant group, that is, CBGs in which the relevant group comprises at least 60% of residents. We replace our dependent variable in our main regression specification with these new measures. The results of these regressions are reported in Tables 6 and 7, respectively.¹⁸

Referring to Table 6, we see inconclusive results regarding visitor race. Depending on our approach to aggregation, we conclude that white visitors may respond more strongly or that Black visitors may do so. Our experiment suggests that some nuance may exist

around gender. As such, our inability to account for visitor gender may play some role in these noisy results.

By contrast, in Table 7, we see a consistent pattern; that is, we see that Black-owned label adoption leads to much larger increases in visits by residents of liberal-leaning CBGs (as compared with right-leaning CBGs). These results are aligned with our experimental findings that personal politics moderate response to the Black-owned label.

5. Discussion and Conclusion

We examine the influence of proprietor social identity signals, such as Black- and woman-owned, on consumer perceptions, evaluations, and demand. Our controlled experiment with a strategically sampled population shows that the Black-owned business label significantly increases subjects' willingness to visit a retailer and their expectations about retailer quality in terms of food, service, and atmosphere. We also demonstrate heterogeneous effects moderated by subjects' political leanings, race, and gender. Our archival analysis of foot traffic supports these findings, revealing increased demand following the adoption of Black- and woman-owned labels with heterogeneity based on neighborhood political leaning.

Our study contributes to the academic literature on the effects of online identity (Taylor et al. 2023) by uncovering distinct underlying mechanisms by which social identity signals impact retailers' performance. The experiment offers causal evidence that the Black-owned label increases consumer willingness to patronize a business and improves perceptions of food, service, and atmosphere quality. This complements the reduction in

Table 6. Effect by Visitor Home CBG Racial Mix (Dependent Variable = Monthly Visits), Black-Verified Panel

DV = Visits from	Est. white visitors (1)	Est. Black visitors (2)	≥50% white CBG (3)	≥50% Black CBG (4)	≥60% white CBG (5)	≥60% Black CBG (6)
Yelp Black-owned	9.647* (3.745)	10.895** (3.343)	10.292* (4.237)	10.572** (3.472)	8.372* (3.704)	10.120** (3.193)
Google Black-owned	6.912*** (1.900)	7.912** (2.694)	6.737** (2.045)	8.176** (2.840)	5.481** (1.789)	6.653* (2.639)
Yelp women-owned	−8.107+ (4.826)	−0.589 (2.941)	−9.488+ (5.545)	0.093 (3.057)	−8.999+ (4.968)	0.015 (2.706)
Google women-owned	−2.124 (2.609)	−1.597 (2.619)	−2.012 (2.929)	−1.570 (2.714)	−1.948 (2.574)	−1.765 (2.582)
Yelp LatinX-owned	−2.435 (6.153)	−4.203 (6.992)	−1.891 (6.990)	−5.430 (8.942)	−1.049 (6.181)	−3.011 (6.877)
Google veteran-led	8.775 (7.822)	1.264 (2.066)	9.325 (8.576)	0.452 (2.126)	9.384 (8.412)	0.303 (1.946)
Google LGBTQ friendly	6.377* (2.711)	1.546 (2.036)	6.844* (2.997)	1.231 (2.094)	5.876* (2.694)	0.592 (1.965)
Number of observations	20,306	20,306	20,306	20,306	20,306	20,306
Adjusted R^2	0.561	0.604	0.568	0.618	0.570	0.620
Adjusted R^2 within	0.007	0.007	0.006	0.007	0.006	0.006
Fixed effects	Place, city-month-year					

Note. Standard errors clustered by retailer.

+ $p < 0.1$; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

search frictions facilitated by these labels on platforms such as Yelp and Google Places. An open question here, which presents an opportunity for future work, pertains to the potential benefits of deploying these labels in online advertising, and potentially targeting ads based on consumer demographics or indicators of political preference.

Of course, our study has some limitations, including the inability to infer the exact dates of label adoption, which leads to measurement error in treatment onset. However, this error is likely to produce conservative estimates of treatment effects as the misclassification

amounts to incorrectly coding untreated panels as treated prior to actual treatment onset (Denteh and Kédagni 2022). Other potential confounding factors include owners' decisions to disclose their social identity and broader marketing efforts. The consistency between our experimental and archival analyses lends credibility to our findings although further experimental interventions in terms of actual consumer behavior could help address these identification concerns.

Extensions of this work could involve incorporating consumer spending data to gain more granular insights

Table 7. Effect by Visitor Home CBG Political Mix (Dependent Variable = Monthly Visits) in the Black-Verified Panel

DV = Visits from	Est. Rep visitors (1)	Est. Dem visitors (2)	≥50% Rep CBG (3)	≥50% Dem CBG (4)	≥60% Rep CBG (5)	≥60% Dem CBG (6)
Yelp Black-owned	5.156** (1.720)	21.841*** (6.559)	1.219* (0.547)	25.882*** (7.798)	0.540* (0.253)	1.219* (0.547)
Google Black-owned	3.591** (1.098)	13.901** (4.951)	2.247** (0.813)	15.436** (5.674)	1.416* (0.643)	2.247** (0.813)
Yelp women-owned	−4.639+ (2.520)	−7.190 (7.051)	−3.645+ (2.040)	−8.045 (8.489)	−1.125+ (0.602)	−3.645+ (2.040)
Google women-owned	−0.737 (1.046)	−3.192 (3.210)	−0.464 (1.129)	−3.414 (3.669)	−0.583 (0.771)	−0.464 (1.129)
Yelp LatinX-owned	−1.608 (2.552)	−7.477 (13.426)	0.605 (1.667)	−9.757 (14.999)	−0.114 (0.690)	0.605 (1.667)
Google veteran-led	3.375 (2.541)	0.169 (3.431)	6.402+ (3.804)	−3.090 (4.361)	5.466+ (3.246)	6.402+ (3.804)
Google LGBTQ friendly	2.724* (1.195)	2.372 (3.005)	2.829* (1.176)	2.126 (3.651)	2.074* (0.914)	2.829* (1.176)
Number of observations	20,306	20,306	20,306	20,306	20,306	20,306
Adjusted R^2	0.710	0.618	0.781	0.611	0.783	0.781
Adjusted R^2 within	0.010	0.010	0.010	0.009	0.009	0.010
Fixed effects	Place, city-month-year					

Notes. Standard errors clustered by retailer. Columns (1) and (2) provide a rough estimate.

+ $p < 0.1$; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

into the nature of incremental visits and examining the influence of other social identity labels, such as LGBTQ+ allyship, on retailer performance and consumer demand. Recent work on the consequences of LGBTQ+ status in online markets (Mejia and Parker 2021) suggests that these related labels may also impact consumer demand, offering potential avenues for future research.

Endnotes

¹ We present examples of the Black-owned business labels from Google Maps (Places) and Yelp in Online Appendix A.

² Thus, it is unlikely that any effects seen would be driven by the salience of the BLM movement, which was most prominent in the summer of 2020 and more likely reflect the longer run influence of these labels.

³ Our findings are not sensitive to the inclusion or omission of these fixed effects.

⁴ We present two exemplary variations for a single business in Online Appendix B: one for our control condition and one containing the Black-owned treatment. The design and flow of the experiment are also articulated with screenshots from the digital survey.

⁵ Similar questions are employed in multidimensional ratings of restaurants at [TripAdvisor.com](https://www.tripadvisor.com). Academically, these dimensions are also shown to matter the most in restaurant settings (Sim et al. 1999).

⁶ We provide visual depictions of covariate imbalance in the unadjusted and adjusted samples in Online Appendix C.

⁷ Wang et al. (2022), in particular, report that marketers' attempts to align themselves with the BLM movement led to negative consumer responses on social media with the effects being moderated by consumers' political leaning. A similar mechanism may be expected to arise here as consumers' responses to social identity labels may depend on how they view the underlying social movements that initially gave rise to the labels. Additionally, there is good reason to believe that responses may be stronger among members of the marginalized groups in which the owner has signaled to be a member based on simple notions of in-group preference (Tajfel and Turner 2004). Recent surveys by the Pew Research Center suggest that Americans' support for both BLM and #MeToo varies by race, gender, political leanings, and age. Generally, whereas Democrats were largely in favor of both causes, Republicans expressed lower support or even opposition. Support for BLM was nearly double among Black respondents compared with white respondents, and that support has not diminished over time, whereas it has faded among other racial groups.

⁸ Note that we can safely omit either label from our estimations without concern for bias because each label is randomly assigned and, thus, orthogonal to all other variables in our regression.

⁹ Here, the reference value is very liberal, coded as zero. This measure is a four-value ordinal scale; thus, it ranges from zero to three in this analysis with three referring to very conservative.

¹⁰ We use point of interest-level (<https://www.safegraph.com/products/places>) and monthly patterns (<https://docs.safegraph.com/docs/monthly-patterns>) data from Safegraph.

¹¹ The choice of these cities is based on them being racially and socioeconomically diverse and offering a wide range of food and beverage establishments, both essential for the generalizability of our results.

¹² Nearly 27% of all locations that we initially sampled were part of a national or global food and beverage chain. Among chain

locations, only 1.1% adopt a Black-owned label, and only 0.07% adopt a women-owned label. We observed null results using this subsample across all of our analyses. We are hesitant to draw strong conclusions from these null results given the sparsity of label adoption. However, results for our main specification, Equation (3), using the sample of chain restaurants are provided in Online Table E2 for transparency.

¹³ Full names, gender, race, high-quality profile images, owner partnership information, and other pertinent information was manually gathered via web searches, inspections of social media profiles, newspaper articles, online directories, and business listing websites. Each coder's inference about owner race and gender was cross-validated against Tzioumis' (2018) demographic data as well as name-based gender and race predictions from the social security database as implemented in the *predictr* package in R. On average, the coder inferences about owner race and gender agreed with the name-based predictions 87% and 89% of the time, respectively.

¹⁴ Note that we do not include city fixed effects in this specification as these are subsumed by our restaurant fixed effects.

¹⁵ See VICE Motherboard: "Uber Eats Lets Anyone Identify as the Owner of a 'Black-Owned' Restaurant."

¹⁶ Table E1 in Online Appendix E replicates Table 5 on the sample, including independent and chain establishments as a robustness check; results remain consistent. Bertrand et al. (2004) raises the concern of multiple periods in DiD estimation introducing autocorrelation; to ensure that this doesn't impact our results, we also estimate the main models using an annual per-retailer aggregated panel in Online Table E3, and the results are comparable.

¹⁷ We use 2020 estimates from the precleaned census data provided by Safegraph (<https://www.safegraph.com/free-data/open-census-data>).

¹⁸ Note that we focus on the Black-owned label in this analysis because we are unable to construct analogous measures of demographic homophily for the woman-owned business label given there is no notion of a male- or female-majority CBG; gender is generally distributed evenly in any given CBG.

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